DATABASE MAP — CURRENT vs TARGET

CURRENT IMPLEMENTED
SQL Server 2022
LongevityDietDb
3 tables: Users / UserProfiles / RefreshTokens
1 DbContext, one SQL database

TARGET ARCHITECTURE (NOT YET IMPLEMENTED)
Six logical service-owned SQL databases
33 canonical logical table types, distributed by owner
No cross-database foreign keys or shared DbContext

Identity SQL
7 entities

Catalog SQL
8 entities

Planning SQL
7 entities

Tracking SQL
7 entities

Recommendation SQL
1 canonical + proposed

Worker SQL
3 canonical + proposed

Cloudflare D1
PROPOSED seventh store

Interpretation: 7 + 8 + 7 + 7 + 1 + 3 = 33 distinct table definitions from the assignment documentation. Outbox / inbox tables are copied to every producer or consumer service as required. Cloudflare D1 is an additional FUTURE proposal and is not part of the current six SQL DBs.

Sources: docs/06-DATABASE-DESIGN.md; docs/assignment/03-DATA-MODEL.md; EF Core LongevityDietDbContext; ADR-005. Exact DDL exists only for the current 3 tables. Future fields are specified/proposed, NOT migrations.

CURRENT SQL — DEPLOYED

LongevityDietDb: physical SQL Server database from EF Core migration. ONLY these 3 tables exist in code as of main 1962735.



RELATIONSHIP & INTEGRITY NOTES

- Users 1 : 1 UserProfiles
- Users 1 : many RefreshTokens

TokenHash (SHA-256), never plaintext in SQL. Refresh token delivered as HttpOnly browser cookie.

TARGET 1/6 — IDENTITY & PROFILE SQL

LongevityIdentityDb — LOGICAL TARGET, not deployed. The Cloudflare D1 design page shows a future RefreshToken storage migration.

Users

```
PK Id : Guid
UQ NormalizedEmail : string(320)
Email : string(320)
PasswordHash : string
DisplayName : string(120)
Role, Status : enums
CreatedAt, UpdatedAt : UTC
```

UserProfiles

```
PK/FK UserId -> Users.Id
BirthYear : int?
TimeZone : string(100)
WakeTime, SleepTime : time?
PreferredMealFrequency : int
FoodPreference : string?
ProfileCompleted : bool
UpdatedAt : UTC
```

RefreshTokens [legacy]

```
PK Id : Guid
FK UserId -> Users.Id
UQ TokenHash : SHA-256
CreatedAt, ExpiresAt : UTC
RevokedAt : UTC?
ReplacedByTokenId : Guid?
Target after Cloudflare cutover: remove
```

Allergies

```
PK Id : Guid
UQ Code : string
Name : string
```

UserAllergies

```
PK (UserId, AllergyId)
UserId -> Users.Id
AllergyId -> Allergies.Id
Severity, Note : optional
```

UserExcludedFoods

```
PK Id : Guid
UserId -> Users.Id
FoodId : Guid? external ID only
FreeText, Reason : string?
CHECK FoodId OR FreeText
```

Reminders

```
PK Id : Guid
UserId -> Users.Id
ReminderType : enum
ScheduleLocalTime : time
DaysOfWeekJson : JSON
IsEnabled : bool
NextDueAtUtc : UTC
```

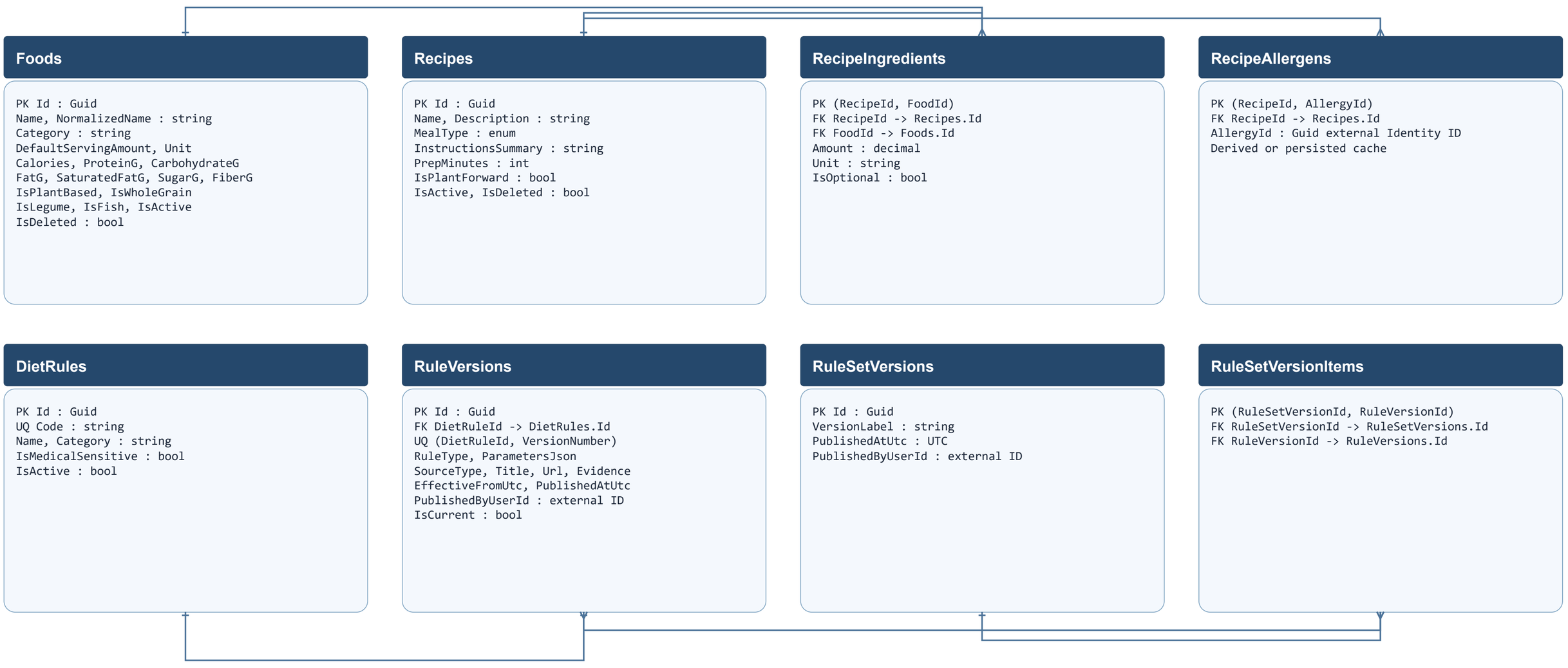
RELATIONSHIP & INTEGRITY NOTES

- Users 1:1 UserProfiles
- Users 1:N UserAllergies, UserExcludedFoods, Reminders
- Allergies 1:N UserAllergies
- RefreshTokens migrate to Cloudflare D1 only after adapter + tests

No SQL foreign key from UserExcludedFoods.FoodId to CatalogDb.Foods. IDs are external references.

TARGET 2/6 — CATALOG & RULES SQL

LongevityCatalogDb — 8 documented logical tables. SQL FK references remain inside Catalog only.



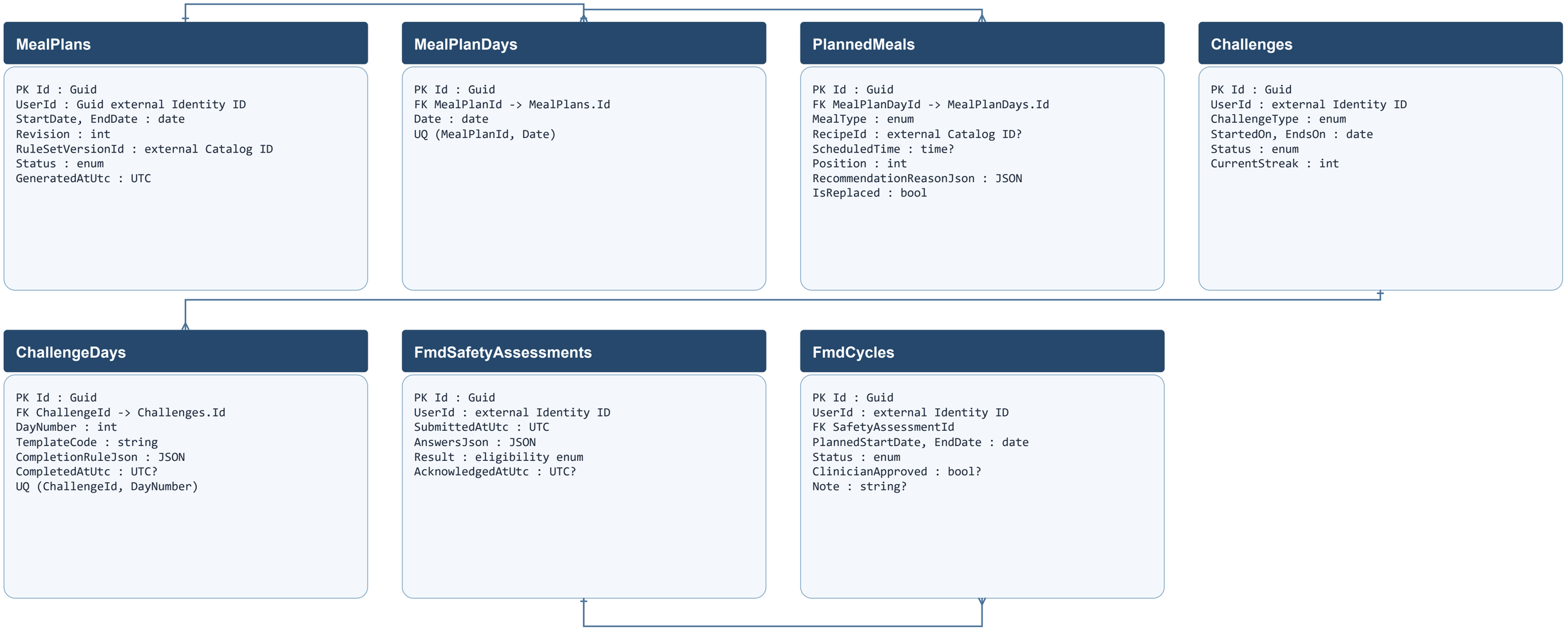
RELATIONSHIP & INTEGRITY NOTES

- Foods 1:N RecipeIngredients
- Recipes 1:N RecipeIngredients and RecipeAllergens
- DietRules 1:N RuleVersions
- RuleSetVersions N:M RuleVersions via RuleSetVersionItems

AllergyId and PublishedByUserId are cross-service stable IDs, NOT foreign keys across databases.

TARGET 3/6 — PLANNING SQL

LongevityPlanningDb — meal plans, challenges and FMD safety; no cross-database joins.



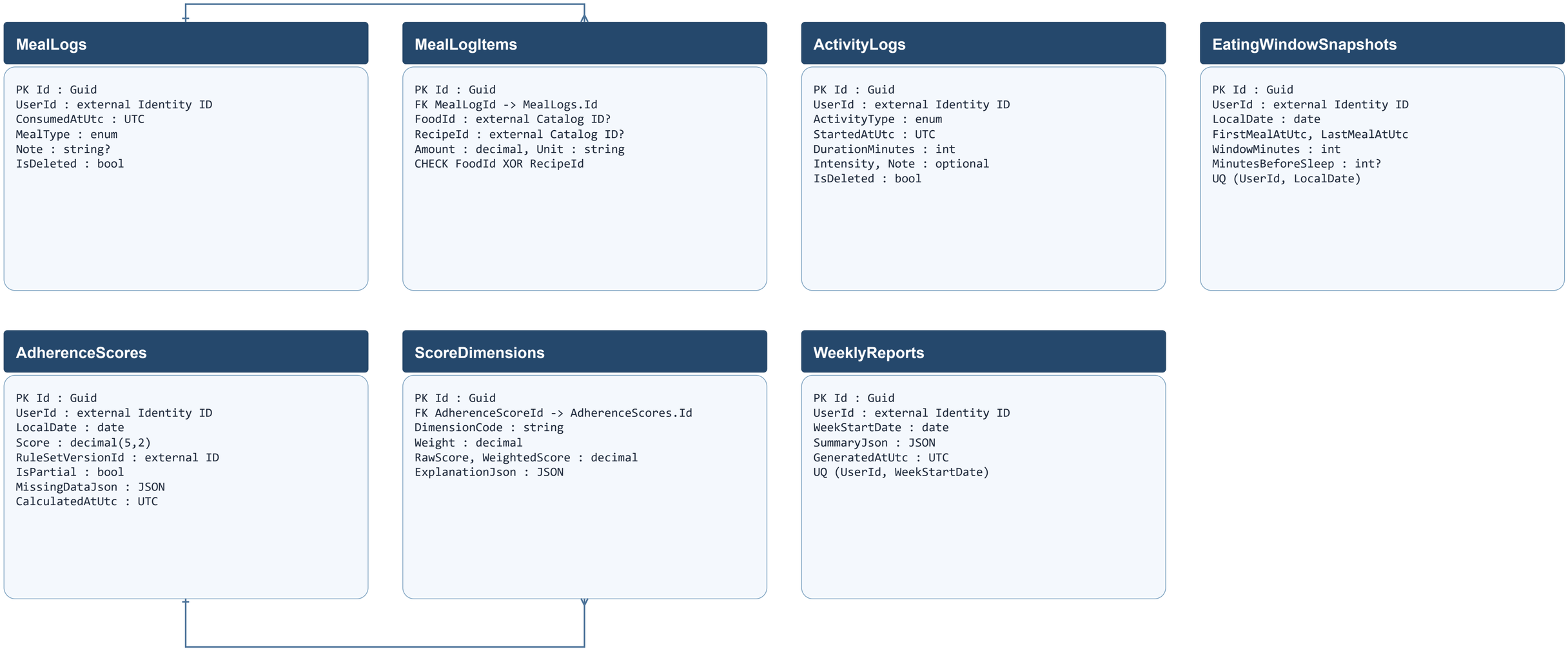
RELATIONSHIP & INTEGRITY NOTES

- MealPlans 1:N MealPlanDays 1:N PlannedMeals
- Challenges 1:N ChallengeDays
- FmdSafetyAssessments 1:N FmdCycles

Planning -> Recommendation is gRPC/HTTP2 :8085. RecipeId, RuleSetVersionId and UserId are external stable IDs.

TARGET 4/6 — TRACKING & PROGRESS SQL

LongevityTrackingDb — logs, adherence, projections; Worker publishes report result rather than writing this DB.



RELATIONSHIP & INTEGRITY NOTES

- MealLogs 1:N MealLogItems
- AdherenceScores 1:N ScoreDimensions
- WeeklyReports arrive from idempotent worker-result events

Idempotency and transactional Outbox local tables should be added per producer/consumer; see technical storage page.

TARGET 5/6 — RECOMMENDATION SQL

LongevityRecommendationDb — documentation defines feedback and conceptual local snapshots; snapshot physical DDL does NOT yet exist.

RecommendationFeedback

```
PK Id : Guid
UserId : external Identity ID
RequestId : Guid
RecipeId : external Catalog ID
FeedbackType : enum
Comment : string?
CreatedAtUtc : UTC
```

CatalogSnapshots [PROPOSED]

```
PK SnapshotId : Guid
SourceCatalogVersion : string
FoodOrRecipeId : Guid
SafePayloadJson : JSON
ReceivedAtUtc : UTC
UQ (Version, SourceId)
```

RuleSnapshots [PROPOSED]

```
PK SnapshotId : Guid
RuleSetVersionId : Guid
RulePayloadJson : JSON
ReceivedAtUtc : UTC
Indexed for active version
```

RankingRequests [PROPOSED]

```
PK RequestId : Guid
UserId : external Identity ID
RuleSetVersionId : Guid
RequestedAtUtc : UTC
Status / CorrelationId
Do NOT persist raw health secrets
```

RELATIONSHIP & INTEGRITY NOTES

- Feedback.RequestId links to request metadata, not mandatory FK until schema is approved
- Catalog and Rule snapshots populated from versioned integration events

Only RecommendationFeedback belongs to the canonical 33-table model; other three boxes are explicitly proposed extensions.

TARGET 6/6 — BACKGROUND WORKER SQL

LongevityWorkerDb — operational ownership. Technical Outbox/ProcessedEvent patterns are cloned locally into publishing/consuming service DBs when needed.

ProcessedEvents

```
PK (EventId, ConsumerName)
EventId : Guid
ConsumerName : string
ProcessedAtUtc : UTC
UQ event + consumer
```

OutboxMessages

```
PK Id/EventId : Guid
EventType : string
PayloadJson : JSON
OccurredAtUtc : UTC
PublishedAtUtc : UTC?
AttemptCount : int
LastError : string?
```

AuditLogs

```
PK Id : Guid
ActorUserId : external Identity ID
Action, EntityType : string
EntityId : Guid
SafeDiffJson : JSON
CorrelationId : string
OccurredAtUtc : UTC
```

WorkerJobs [PROPOSED]

```
PK JobId : Guid
JobType, Status : string
DueAtUtc, LastAttemptAtUtc
Attempts / CorrelationId
Completion metadata
```

DeadLetterEvents [PROPOSED]

```
PK EventId : Guid
ReasonCode, SafePayloadRef
FailedAtUtc : UTC
RetryCount : int
ResolutionStatus : string
```

RELATIONSHIP & INTEGRITY NOTES

- ProcessedEvents idempotent per consumer
- OutboxMessages owned by each producer, never one central database
- AuditLogs domain-owned; centralized ops events must not write foreign DB

WorkerJobs and DeadLetterEvents are design extensions. Worker must NEVER poll other services' Outbox directly.

CLOUDFLARE D1 — FUTURE TOKEN DATABASE

PROPOSAL ONLY — Cloudflare account/binding not configured. Move refresh-session HASH metadata, not raw refresh tokens, using a Cloudflare Worker as sole API.

RefreshSessionRecords [PROPOSED]

```
PK TokenId : UUID
UQ TokenHash : SHA-256 hex
UserId : stable Identity ID
CreatedAtUtc, ExpiresAtUtc
RevokedAtUtc : UTC?
ReplacedByTokenId : UUID?
FamilyId : UUID (rotation lineage)
DeviceLabel : optional, non-sensitive
```

SessionReplayAudit [PROPOSED]

```
PK AuditId : UUID
TokenId : UUID
DetectedAtUtc : UTC
Action : revoke family
CorrelationId : string
```

Cloudflare Worker API [NOT A TABLE]

```
POST /sessions/issue
POST /sessions/rotate
POST /sessions/revoke
Authenticate service to service
D1 binding is private
Do not publish raw tokens in logs
```

RELATIONSHIP & INTEGRITY NOTES

- Identity API -> authenticated Cloudflare Worker -> D1 (private binding)
- Use atomic D1 transaction / conditional rotate to prevent replay and race
- Old SQL RefreshTokens must remain until migration has completed

Cloudflare Workers Secrets store API credentials/signing keys; D1 stores refresh-session hashes and expiry, not raw tokens.